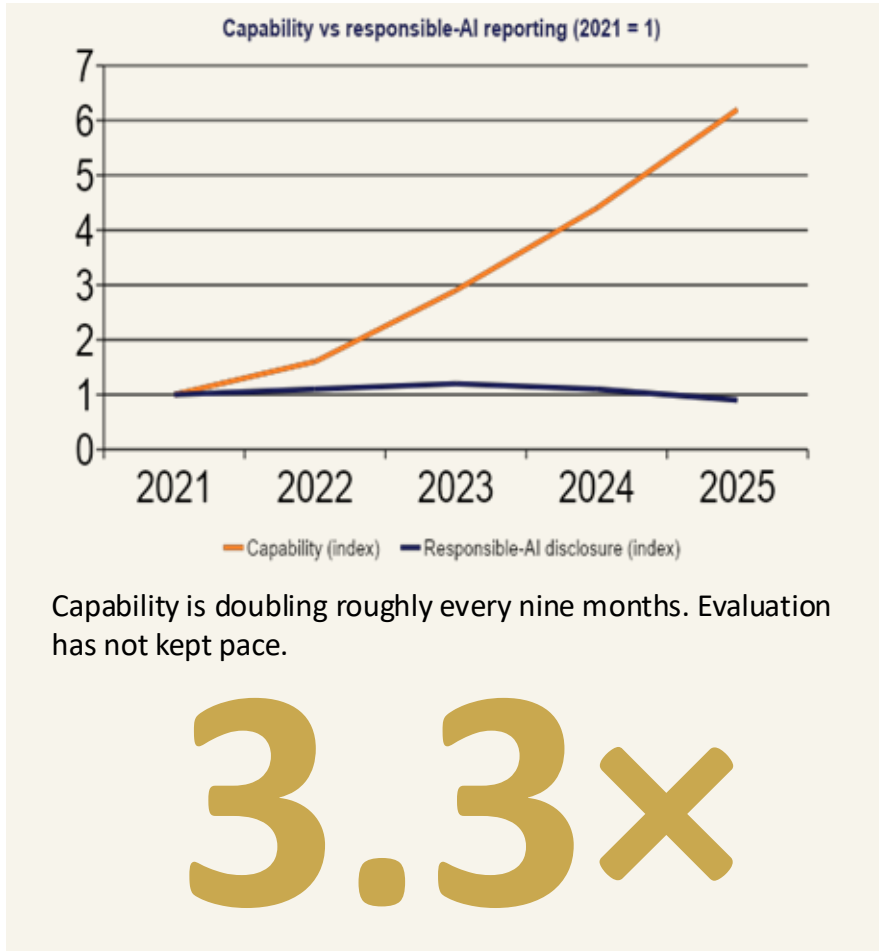


Eclipse PanEval: Architecting Neutral AI Evaluation for the Era of the EU AI Act

Michael Berns
Head of AI Eclipse Foundation



AI capability is running ahead of responsible reporting



The trust gap is structural

- Frontier developers disclose less of their methodology each year.
- Public benchmarks saturate faster than they are refreshed.
- Independent tests do not always confirm developer claims.
- Public trust in AI governance remains fragmented.

If capability scales 3.3x per year, evaluation has to scale with it — openly, neutrally, and by default.

Source: Stanford HAI, AI Index Report 2026.

Three numbers that frame the trust gap

90%+

of notable 2025 frontier models came from industry — and disclosed less of their methodology.

50.1%

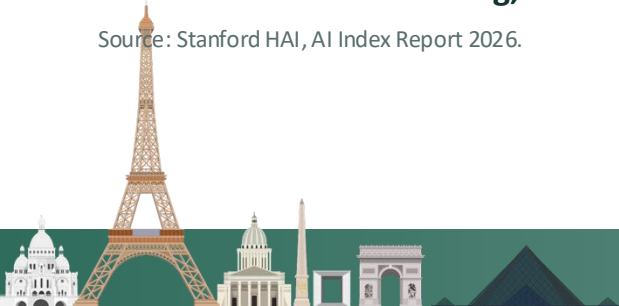
accuracy when leading models are asked to read an analog clock.

31%

U.S. public trust that government can regulate AI effectively.

Benchmarks are saturating, frontier developers disclose less, and independent testing rarely confirms what developers report.

Source: Stanford HAI, AI Index Report 2026.



01

The regulatory forcing function

The EU AI Act turns benchmarking from good practice into legal obligation.



A risk-based regime, fully enforced by August 2026



Prohibited	Unacceptable risk uses are banned outright (Article 5).
High-risk	Conformity assessment, logging, human oversight (Article 6).
Limited risk	Transparency duties for users and deployers.
Minimal risk	Voluntary codes — most systems sit here today.

Sources: bitkom Umsetzungsleitfaden KI (2024); EY AI Act — GenAI and Agentic, Part 1 (March 2026).



What the AI Act asks of providers of general-purpose AI

Technical documentation



Training data summaries, architecture and compute descriptions — maintained and available to authorities.

Model evaluation



Adversarial testing and systemic risk assessment for high-capability models, documented and reproducible.

Safety and cybersecurity



Incident reporting, cyber-resilience aligned with the Cyber Resilience Act, vulnerability handling across the lifecycle.

Copyright and provenance



Policies to respect EU copyright law including opt-outs, and to disclose training data sources at a summary level.

Evaluation is the connective tissue — it turns each duty into something observable, repeatable, and audit-ready.



No open framework today does all three things at once



Aligned with EU conformity

Most benchmarks were designed for English-first, U.S.-centric research settings — not for the Article 95 obligations the EU AI Act now requires.



Covering safety and cyber together

Safety, bias, and robustness usually live in one place. Cybersecurity and vulnerability handling live in another. The AI Act treats them as one job.



Vendor-neutral, community-governed

Leaderboards and evaluation suites are often controlled by a single developer. Governance is the differentiator, not the metric.

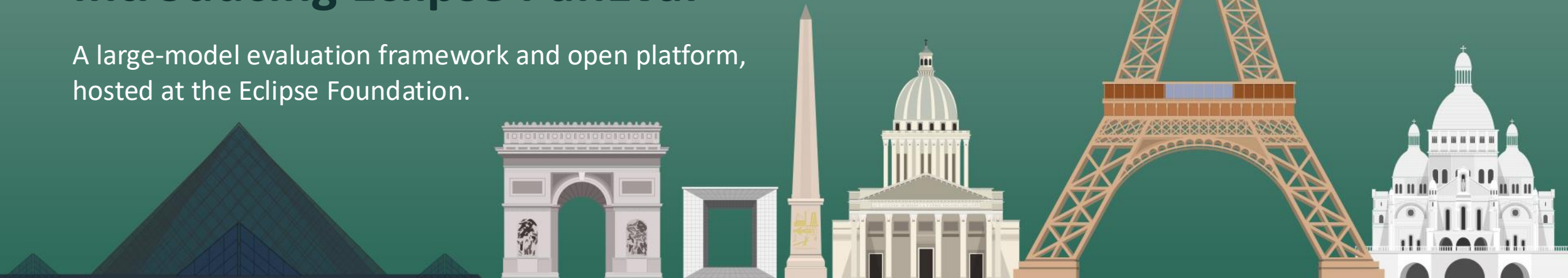
Eclipse PanEval is designed to close this gap — open source, vendor neutral, and built for the EU regime.



02

Introducing Eclipse PanEval

A large-model evaluation framework and open platform,
hosted at the Eclipse Foundation.



Scientific, impartial, open — built for AI reliability

What it is

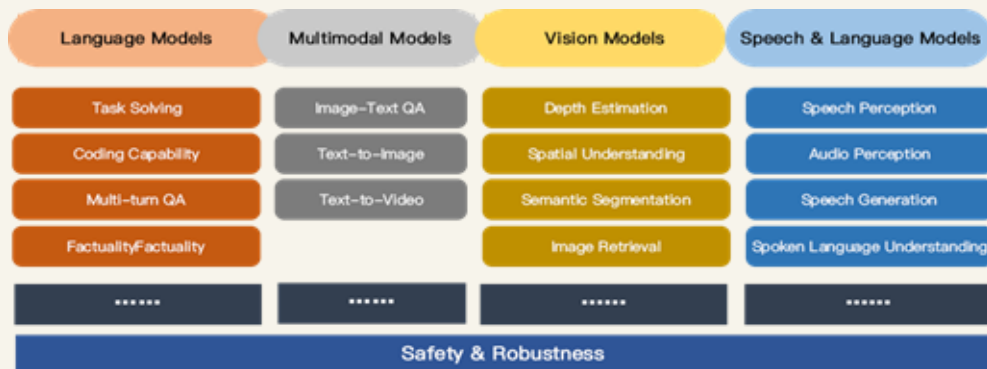
An open source framework and platform to evaluate foundation models — covering performance, safety, robustness, and bias.

Why it exists

To give Europe a verifiable, vendor-neutral path to AI reliability that maps to the AI Act's evaluation obligations.

How it is licensed

Apache 2.0 — compatible with enterprise adoption and downstream commercial use.



Origin

Builds on FlagEval — five years of evaluation work at the Beijing Academy of Artificial Intelligence (2020–2025).

Home

Hosted at the Eclipse Foundation from 2026.

Coverage

Four model categories — language, multimodal, vision, speech. More than 40 evaluation tasks at launch.

Standards

Active contribution to IEEE P3419 — international standard for large-model evaluation.



Capacity × Task × Metrics — mapped to the obligations that matter

THE THREE-DIMENSIONAL MODEL

Capacity

Scope of model capabilities — what a model is asked to do.

Task

Benchmarks, datasets, adversarial tests that probe each capability.

Metrics

Accuracy, robustness, safety, fairness — multi-perspective scoring.

MAPPED TO THE AI ACT

Article 15 · Accuracy, robustness, cybersecurity

42 evaluation tasks across four modalities; Video-SafetyBench attack-success scoring; ROME and TRUE benchmarks for robustness.

Article 53 · GPAI transparency

The Capacity–Task–Metrics framework maps directly to Annex XI evaluation evidence; structured JSON records.

Article 55 · Systemic risk

paneval-serving API for AI Office workflows; versioned task configs for a full audit trail.

PanEval supplies the evaluation evidence the AI Act asks for — it does not generate compliance documents itself.



Accuracy, robustness and cybersecurity — mapped to PanEval capabilities

Accuracy Art.15(1)–(3)

- ▶ **42 evaluation tasks** across NLP, CV, Multimodal, Audio
- ▶ **13 evaluator types** — exact-match to semantic LLM-judge
- ▶ **LLM-as-Judge** for automated subjective evaluation
- ▶ **Multi-inference aggregation** — cross-run consistency scoring
- ▶ **Structured JSON reports** with per-task breakdowns, versioned

Robustness Art.15(4)

- ▶ **ROME benchmark** — 7 dimensions × 7 matching strategies
- ▶ **TRUE benchmark** — input perturbation resilience via edit distance
- ▶ **HallusionBench** — factual grounding under adversarial framing
- ▶ **Cross-run stability** — multi-inference aggregation surfaces stochastic failures

Cybersecurity Art.15(5)

Operational today

- ▶ **Video-SafetyBench** — 5-level RJScore via Qwen 72B judge
- ▶ **Attack Success Rate** — per-scenario unsafe ratio
- ▶ **Safety evaluation API** — standardised endpoint in production

Phase 2 roadmap

- Prompt injection detection
- Data poisoning benchmarks
- Model extraction resistance

The PanEval Solution — enabling GPAI provider transparency obligations

The Requirement: *Art.53(1)(a) maintain up-to-date technical documentation including training/testing process and evaluation results. Art.53(1)(b) provide downstream providers with information to understand capabilities and limitations.*

Art.53(1)(a) Technical Documentation

How PanEval supports Annex XI evaluation evidence

- ▶ **Capacity–Task–Metrics framework** maps directly to Annex XI "evaluation strategies, testing and validation measures, metrics used"
- ▶ **42 tasks × 13 evaluators** produce structured JSON evaluation records — machine-readable, versioned per model release
- ▶ **Reproducible configs** — every evaluation run is deterministic (seeds, prompts, thresholds recorded)
- ▶ **Longitudinal tracking** — version-stamped results enable drift detection across model updates

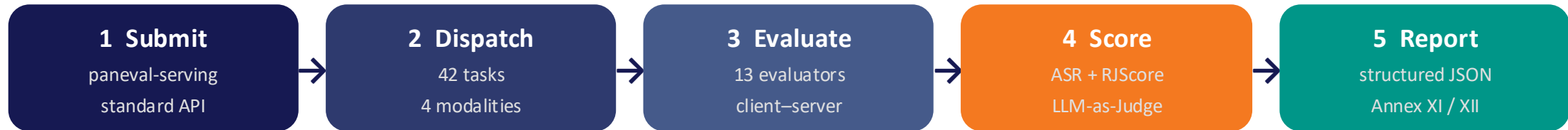
Art.53(1)(b) Downstream Provider Information

How PanEval enables capability & limitation disclosure

- ▶ **paneval-serving API** — standardised model submission + evaluation endpoint; downstream providers query results programmatically
- ▶ **Structured capability profile** — per-task scores + known limitations = machine-readable disclosure document
- ▶ **Arena comparison** — 21 LLMs + 33 multimodal models in head-to-head; downstream can benchmark relative positioning
- ▶ **Open leaderboard** — transparent, public, continuously updated capability rankings

PanEval provides the evaluation evidence that documentation must contain — it does not generate compliance documents itself. Art.53(1)(c) copyright and Art.53(1)(d) training data summary: covered via dataset provenance tracking and IEEE P3419 aligned methodology.

The PanEval Solution — evaluation infrastructure for systemic risk



Art.55(1)(a) Adversarial Testing

- ▶ **Video-SafetyBench** — 5-level RJScore with Qwen 72B automated judge
- ▶ **Attack Success Rate** — per-scenario unsafe ratio, the metric Art.15(5) demands
- ▶ **HallusionBench** — factual grounding under adversarial framing
- ▶ **ROME** — multi-dimensional cognitive robustness stress test

Art.55(1)(b) Systemic Risk Assessment

- ▶ **Client-server architecture** — remote evaluation without model transfer; fits AI Office workflow
- ▶ **42-task capability mapping** — comprehensive risk surface analysis across 4 modalities
- ▶ **Multi-inference stability** — stochastic failure mode detection via repeated sampling
- ▶ **343+ models evaluated** — proven scalability for high-capability GPAI models

Art.55(1)(c) Incident Tracking

- ▶ **Versioned task configs** — full audit trail for every evaluation run
- ▶ **Longitudinal comparison** — detect capability regression across releases
- ▶ **Apache 2.0 codebase** — fully inspectable by regulators and third parties

Art.51 Classification

Systemic risk if training compute exceeds 10^{25} FLOPs or by Commission designation. PanEval provides the evaluation layer — tasks, metrics, and reporting infrastructure. Model training, deployment, and serving remain the provider's responsibility.

The AI Office may request or conduct these evaluations directly (Art.55 final para). PanEval is designed to serve as that infrastructure — open, auditable, vendor-neutral.

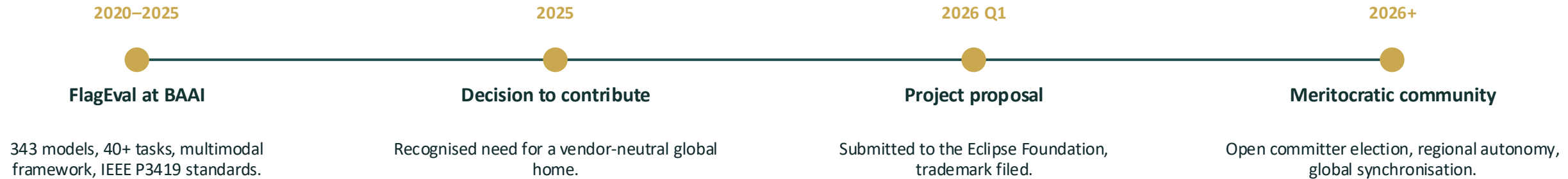
03

Meritocratic stewardship

From a foundational contribution to a globally governed, community-led project.



From FlagEval to a dual-sync, globally governed open source project



CHINA · BAAI track

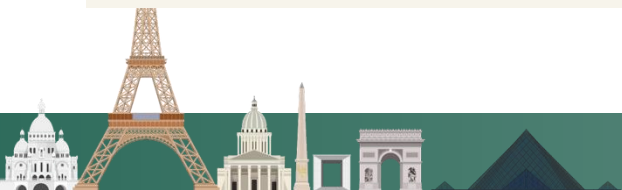
- FlagEval codebase and Chinese leaderboard (343 models).
- Chinese-specific datasets and IEEE P3419 standards.
- China regulatory compliance and FlagEval Arena.

EUROPE · Eclipse track

- Eclipse PanEval — governed release.
- EU-compliant datasets, AI Act and CRA alignment.
- Meritocratic committer process and Eclipse IP due diligence.

**DUAL
SYNC**

Shared core. Independent evolution. Synchronised releases.



Three outcomes that land with adopters, contributors, and regulators

01

Regional de-risking

Each side operates under its own governance. Regulatory or geopolitical shocks in one region do not propagate into the other.

02

Preserved innovation

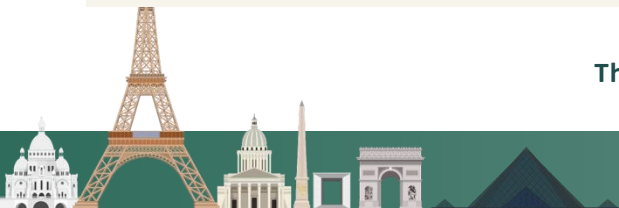
A shared core prevents fragmentation. Improvements anywhere benefit everywhere, on a synchronised release cadence.

03

Compliance by design

The European track is EU-aligned from day one — not retrofitted after deployment. That is what auditors and procurement teams now ask for.

These three outcomes are what auditors, procurement teams, and open source contributors say they need next.



Benchmarking trust is a community job.

Join us in building Eclipse PanEval — openly.

How to contribute

- Propose benchmarks, datasets, and metrics on the project repo.
- Join the working group as a Strategic, Contributing, or Committer member.
- Help suggest evidence templates.

<https://projects.eclipse.org/projects/technology.paneval>

<https://github.com/eclipse-paneval>

Michael Berns · Head of AI, Eclipse Foundation

<https://www.linkedin.com/in/aithoughtleader/>

michael.berns@eclipse-foundation.org

